Yan Ma

600+ citations, 1.3k+ GitHub stars across research projects

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Research Interests

My research aims to build multimodal agents for partially observed environments, especially how they perceive, reason, act, and predict over long-horizon tasks. So far, my work has mainly studied how RL changes multimodal models, including training dynamics, the joint improvement of reasoning and perception, and agentic vision tool use. Recently, I am interested in predictive multimodal learning from video, especially how video data can be cleaned, filtered, and annotated to better support the anticipation of future observations and action outcomes.

Education

- **Ph.D. Student, Fudan University** (Advisors: Prof. Pengfei Liu and Prof. Yu Qiao) *2023.09 – 2027.06*
- **Master in Computer Application Technology, Fudan University** *2020.09 – 2023.06*
- **Bachelor in Computer Science, Dalian University of Technology** (GPA: 4.076/5) *2016.09 – 2020.06*

Experience

- **Shanghai Innovation Institute** *(2026.03 – 2026.06)*: leading a large-scale audio-video data processing pipeline
- **MiniMax & HailuoAI** *(2025.01 – 2026.02)*: worked on multimodal foundation models for visual reasoning and vision tool use in the M-series, and image caption systems for Hailuo video generation
- **Shanghai AI Laboratory** *(2023.07 – 2025.01)*: worked on text generation and unified multimodal model training

Publications

- **What Does Vision Tool-Use Reinforcement Learning Really Learn? Disentangling Tool-Induced and Intrinsic Effects for Crop-and-Zoom** [][]. *arXiv:2602.01334* [ICML 2026 under review]

  Yan Ma†, Weiyu Zhang†, Tianle Li, Linge Du, Xuyang Shen, Pengfei Liu

  TLDR: We show current vision tool-use RL gains are driven mainly by intrinsic capability improvement and reduced tool-induced harm, not true tool-based correction.
- **One RL to See Them All: Visual Triple Unified Reinforcement Learning** [][]. *arXiv:2505.18129* [COLM 2026 under review]

  Yan Ma†, Linge Du†, Xuyang Shen†, Shaoxiang Chen, Pengfei Li, Qibing Ren, Lizhuang Ma, Yuchao Dai, Pengfei Liu, Junjie Yan

  TLDR: We build a unified RL system (V-Triune) for jointly training visual reasoning and perception in VLMs, yielding broad gains across both task types.
- **Rethinking RL Scaling for Vision Language Models: A Transparent, From-Scratch Framework and Comprehensive Evaluation Scheme** [][]. *arXiv:2504.02587*

  Yan Ma, Steffi Chern, Xuyang Shen, Yiran Zhong, Pengfei Liu

  TLDR: We introduce a transparent, reproducible RL pipeline and evaluation scheme for VLMs, and show RL consistently improves generalization over SFT with key training-dynamics findings.
- **ANOLE: An Open, Autoregressive, Native Large Multimodal Models for Interleaved Image-Text Generation** [][][]. *arXiv:2407.06135; extended version accepted to ICLR 2025 Blog Track*

  Ethan Chern*, Jiadi Su*, Yan Ma*, Pengfei Liu

  TLDR: We present an open native autoregressive multimodal model for coherent interleaved image-text generation, with efficient fine-tuning from Chameleon.
- **MoPS: Modular Story Premise Synthesis for Open-Ended Automatic Story Generation** [][]. *ACL 2024*

  Yan Ma, Yu Qiao, Pengfei Liu
- **Weak-to-Strong Reasoning** [][]. *EMNLP Findings 2024*

  Yuqing Yang, Yan Ma, Pengfei Liu
- **OlympicArena: Benchmarking Multi-discipline Cognitive Reasoning for Superintelligent AI** [][]. *NeurIPS Dataset & Benchmark Track 2024*

  Zhen Huang, Zengzhi Wang, Shijie Xia, ..., Yan Ma, ..., Shaoting Zhang, Dahua Lin, Yu Qiao, Pengfei Liu

Research Stack

- Verl, Transformers, Ray, FSDP, vLLM, SGLang, Jax, FFmpeg, Codex, Git, NeoVim, Tmux